

AE 6114 - Fundamentals of Solid Mechanics
Fall 2024

Homework 2: Kinematics

Problem 1

Given the following motion in rectangular Cartesian coordinate

$$\boldsymbol{\chi} = 3X_3\mathbf{e}_1 - X_1\mathbf{e}_2 - 2X_2\mathbf{e}_3,$$

determine the following:

- a) The deformation gradient $\mathbf{F}$.
- b) The right Cauchy-Green tensor $\mathbf{C}$ and the right stretch tensor $\mathbf{U}$.
- c) The rotation tensor $\mathbf{R} = \mathbf{F}\mathbf{U}^{-1}$.
- d) The Green strain tensor $\mathbf{E}$.
- e) The ratio of deformed volume to initial volume.

Problem 2 - Old Quiz 1 Question!

Consider a cube of material of length L , the body occupies the space $\mathcal{B} = \{(X_1, X_2, X_3) : 0 \leq X_1 \leq L, 0 \leq X_2 \leq L, 0 \leq X_3 \leq L\}$ in the reference configuration.

The body undergoes a motion defined by

$$\chi(\mathbf{X}) = \left(\frac{a}{L} X_1 + \gamma X_2 \right) \mathbf{e}_1 + X_2 \mathbf{e}_2 + X_3 \mathbf{e}_3$$

where a and γ are given constants.

- (a) Compute the deformation gradient $\mathbf{F}$, the Right Cauchy-Green tensor $\mathbf{C}$ and the Green strain tensor $\mathbf{E}$.
- (b) Compute the displacement field $\mathbf{u}$ and the infinitesimal strain tensor $\boldsymbol{\epsilon}$.
- (c) Using **both** finite and infinitesimal deformations, calculate the stretch λ in for fibers aligned with the $\mathbf{e}_1$ -direction.
- (d) Using **both** finite and infinitesimal deformations, calculate the change in angle θ between two fibers oriented in the $\mathbf{e}_1$ and $\mathbf{e}_2$ directions.
- (e) Under what conditions is the infinitesimal strain $\boldsymbol{\epsilon}$ an appropriate measure of strain for this motion? For these conditions, show that $\mathbf{E}$ reduces to $\boldsymbol{\epsilon}$ and that your stretch and angle measurements also coincide.

Problem 3

Envision a rectangular bar stretched uniformly in the $\mathbf{e}_1$ -direction. This deformation results in a **constant** strain field given by

$$[\epsilon] = \begin{bmatrix} C_1 & 0 & 0 \\ 0 & -C_2 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

where C_1 and C_2 are constants. Assume infinitesimal deformations throughout this problem. Further, assume that the displacement field depends only on X_1 and X_2 .

- (a) Integrate the strain-displacement relations to determine the displacement components (u_1, u_2, u_3) and identify any rigid-body motion terms in your displacement field.
- (b) Show that the displacement field $\mathbf{u}$ found in part (a) above can be written in the form

$$\mathbf{u} = C_1 X_1 \mathbf{e}_1 - C_2 X_2 \mathbf{e}_2 + \mathbf{A}\mathbf{X} + \mathbf{b}$$

where $\mathbf{A}$ is a skew tensor and $\mathbf{b}$ is a constant vector. Finally, show that the infinitesimal rotation $\boldsymbol{\omega} = \text{skw}(\nabla \mathbf{u}) = \mathbf{A}$.

Problem 4

Consider a body occupying a cylinder of radius R and length L , $\Omega = \{(X_1, X_2, X_3) : X_1^2 + X_2^2 < R, 0 < X_3 < L\}$ in the reference configuration. It undergoes a deformation:

$$\begin{aligned}x_1 &= X_1 \cos(\tau X_3) - X_2 \sin(\tau X_3) \\x_2 &= X_1 \sin(\tau X_3) + X_2 \cos(\tau X_3) \\x_3 &= X_3\end{aligned}$$

where τ is a constant parameter with unit $[1/\text{length}]$.

1. Describe the deformation. What is the meaning of τ ?
2. Find the deformation gradient $\mathbf{F}$ and compute the right Cauchy-Green stretch tensor $\mathbf{C} = \mathbf{F}^T \mathbf{F}$.
3. Calculate the stretch λ for a fiber that, in the reference configuration, is oriented parallel to the plane defined by the basis vectors $\{\mathbf{e}_1, \mathbf{e}_2\}$.
4. Calculate the local change in area for differential elements of oriented area with normal along the $\mathbf{e}_3$ basis vector.
5. Calculate the local change of differential volume.

Problem 5

Consider the change in basis shown in Fig. 1. Here, the reference coordinate frame $\{\mathbf{e}_i\}$ is rotated about the $\mathbf{e}_3$ -axis about an angle θ to yield a new coordinate frame $\mathbf{e}_i^*$. Note this is a simple two-dimensional rotation of the coordinate frame such that the $\mathbf{e}_3$ and $\mathbf{e}_3^*$ directions remain aligned.

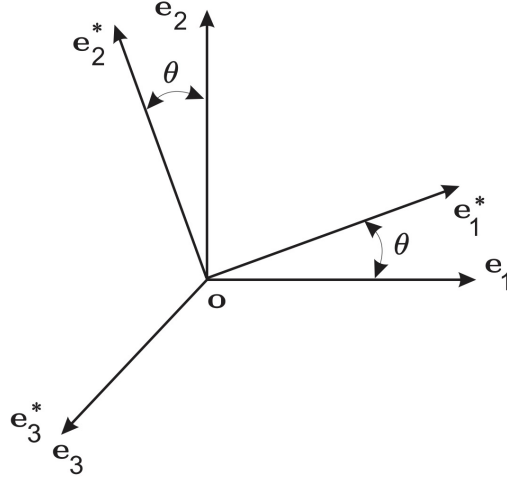


Figure 1: Change in Basis

- For the given transformation, derive the rotation tensor $\mathbf{Q}$.
- Show that the tensor $\mathbf{Q}$ you derived satisfied $\mathbf{Q}\mathbf{Q}^\top = \mathbf{1}$ and $\det(\mathbf{Q}) = 1$.
- Let $\mathbf{A}$ be a **symmetric** tensor. Assume in the reference $\mathbf{e}_i$ frame the tensor $\mathbf{A}$ is spherical. That is in the reference frame it's components are given by

$$[\mathbf{A}] = \begin{bmatrix} A_{11} & 0 & 0 \\ 0 & A_{22} & 0 \\ 0 & 0 & A_{33} \end{bmatrix}$$

Using the transformation rule for tensors, find the components of $\mathbf{A}^*$ as functions of the components of $\mathbf{A}$ and the angle θ .

- Let $A_{11} = 4$ and $A_{22} = 2$. Make a plot of A_{11}^* in the horizontal axis versus A_{12}^* in the vertical axis for $\theta \in [0, 2\pi]$. **Hint:** the result should be a circle, commonly known as the Mohr's circle!

Problem 6

Consider the simple symmetric two-dimensional tensor $\mathbf{S}$ with components

$$[\mathbf{S}] = \begin{bmatrix} 4 & -1 \\ -1 & 4 \end{bmatrix}$$

expressed in a cartesian reference frame $\mathbf{e}_i$.

- a) Find the eigenvalues, λ_i and eigenvectors, $\mathbf{v}_i$ of $\mathbf{S}$. **Show your work!**
- b) Find the rotation tensor $\mathbf{Q}$ that maps from the original reference frame of $\mathbf{S}$ to the eigenspace. That is solve for $Q_{ij} = \mathbf{v}_i \cdot \mathbf{e}_j$.

Show that your matrix $\mathbf{Q}$ is correct by computing $\mathbf{S}^* = \mathbf{Q}\mathbf{S}\mathbf{Q}^\top$. Here $\mathbf{S}^*$ should be a spherical tensor containing the previously found eigenvalues.

- c) Compute $\mathbf{A} = \ln(\mathbf{S})$ the operation of which is defined as

$$\ln(\mathbf{S}) = \sum_{i=1}^3 \ln(\lambda_i) \mathbf{v}_i \otimes \mathbf{v}_i$$

with λ_i the eigenvalues of $\mathbf{S}$ and $\mathbf{v}_i$ the corresponding eigenvectors.

Problem 7

Consider a right Cauchy-Green stretch tensor $\mathbf{C}$. It has a spectral representation $\mathbf{C} = \sum_{i=1}^3 \lambda_i^2 \mathbf{v}_i \otimes \mathbf{v}_i$, where $\lambda_i > 0$ are the principal stretches, $|\mathbf{v}_i| = 1$, and $\mathbf{v}_i$ are mutually orthogonal. Let $\lambda_1 = \lambda_{\max}$, $\lambda_2 = \lambda_{\min}$, and $\lambda_{\max} > \lambda_3 > \lambda_{\min}$:

1. Show that the stretch $\lambda_{\max}$ in the direction $\mathbf{v}_1$ is larger than the stretch λ_N in any other direction $\mathbf{N}$.
2. Show that the stretch $\lambda_{\min}$ in the direction $\mathbf{v}_2$ is smaller than the stretch λ_N in any other direction $\mathbf{N}$.